Activity - 3

Analytical Problem Solving

① Sol

Given $G(x) = 24$

$G(y) = 18$

To find 1. Gradient magnitude

2. Gradient direction

1. Gradient magnitude:

$$\boxed{G = \sqrt{G_x^2 + G_y^2}}$$

$$= \sqrt{(24)^2 + (18)^2}$$

$$= \sqrt{576 + 324}$$

$$= \sqrt{900}$$

$$\boxed{G = 30}$$

2. Gradient direction:

$$\boxed{\theta = \tan^{-1}\left(\frac{G_y}{G_x}\right)}$$

$$\theta = \tan^{-1}\left(\frac{18}{24}\right)$$

$$\theta = \tan^{-1}(0.75)$$

$$\boxed{\theta = 36.870}$$

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② Sol.

Given:

Non - Maximum Suppression

Pixel :	P_1	P_2	P_3	P_4	P_5
Magnitude :	25	56	90	60	30

P_1 : 80 < P_3 → 0

P_2 : 58 < P_3 → 0

P_3 : 90 > P_2

P_4 : 60 < P_3 → 0

P_5 : 30 < P_6 → 0

Non Maximum Suppression

pixel	P_1	P_2	P_3	P_4	P_5
NMS	0	0	90	0	0

∴ P_3 = 90 is retained.

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③ Sol

Given: High Threshold = 100
Low threshold = 50

The gradient magnitudes are

Pixel :	P_1	P_2	P_3	P_4	P_5	P_6
magnitude :	135	90	45	120	70	20

$M \geq 100 \longrightarrow$ strong

$50 \leq M < 100 \longrightarrow$ weak

$M < 50 \longrightarrow$ Non-edge

Pixel	Magnitude	Classification
P_1	135	strong
P_2	90	weak
P_3	45	non-edge
P_4	120	strong
P_5	70	weak
P_6	20	non-edge

4b Sol

Given

Pixel	P_1	P_2	P_3	P_4	P_5	P_6	P_7
magnitude	130	95	40	115	65	25	105

High threshold $= 100$
Low threshold $= 50$

<u>Strong edge</u> $m \geq 100$

$P_1 = 130$, $P_4 = 115$, $P_7 = 105$

<u>Weak edge</u> : $50 \leq m < 100$

$P_2 = 95$, $P_5 = 65$

<u>Non-edge</u> : $M < 50$

$P_3 = 40$, $P_6 = 25$

<u>Hysteresis</u>:

$P_2 \rightarrow P_1$ (retained)

$P_5 \neq$ edge. suppressed

$\therefore$ Final edge map is P_1, P_2, P_4, P_7

⑤ Hough Circle Transform

$$r = \sqrt{(x-a)^2 + (y-b)^2}$$

1) edge points

$P_1 = (5, 4)$

$P_2 = (7, 6)$

$C_1 = (2, 4)$

$C_2 = (4, 3)$

For $C_1 = (2, 4)$

P_1 and C_1 : $r = \sqrt{(5-2)^2 + (4-4)^2}$

$$= \sqrt{3^2 + (0)^2}$$

$$= \sqrt{9}$$

$$\boxed{r = 3}$$

P_2 and C_2 : $r = \sqrt{(7-2)^2 + (6-4)^2}$

$$= \sqrt{25 + 4}$$

$$= \sqrt{29}$$

$$\boxed{r \approx 5.39}$$

P_1 and C_2 : $r = \sqrt{(5-4)^2 + (4-3)^2}$

$$= \sqrt{1+1}$$

$$= \sqrt{2}$$

$$\boxed{r \approx 1.41}$$

P_2 and C_2 : $r = \sqrt{(7-4)^2 + (6-3)^2}$

$$= \sqrt{9+9}$$

$$= \sqrt{18}$$

$$\boxed{r \approx 4.24}$$

Calculate the radius

Given $r = (8, 6)$

$c = (4, 3)$

$r = \sqrt{(x-a)^2 + (y-b)^2}$

$= \sqrt{(8-4)^2 + (6-3)^2}$

$= \sqrt{4^2 + 3^2}$

$= \sqrt{16+9}$

$= \sqrt{25}$

$\boxed{r = 5}$

c) Calculate the radius.

$C = (5, 4)$

$P = (8, 8)$

$r = \sqrt{(8-5)^2 + (8-4)^2}$

$= \sqrt{3^2 + 4^2}$

$= \sqrt{9+16}$

$= \sqrt{25}$

$\boxed{r = 5}$